

SQL Commands & Constraints

Quick Study Notes • DBMS Exam & Interview Prep

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SQL (Structured Query Language) is used to communicate with and manipulate structured data stored in Relational Database Management Systems (RDBMS), primarily in the form of tables.

1. DDL – DATA DEFINITION LANGUAGE

DDL → Structure

Used to define or modify the structure/schema of database objects.

Command	Purpose
CREATE	Creates a database object or table
ALTER	Modifies the existing structure of a table/database
DROP	Removes the structure and all its data permanently
TRUNCATE	Removes all records while retaining the table structure
RENAME	Changes the name of an existing database object

2. DML – DATA MANIPULATION LANGUAGE

DML → Data

Used to manipulate and manage data within existing tables.

Command	Purpose
INSERT	Adds new data records into a table
UPDATE	Modifies existing data values within a table
DELETE	Removes specific or all records from a table

Note: **SELECT** is often classified separately as **DQL (Data Query Language)**, though some academic sources group it with DML.

3. DQL – DATA QUERY LANGUAGE

DQL → Query / Retrieve

Command	Purpose
SELECT	Retrieves data records from one or more tables

4. DCL – CONTROL

Manages permissions and privileges.

GRANT	Gives user access privileges
REVOKE	Takes back user privileges

5. TCL – TRANSACTION

Manages database transactions.

COMMIT	Saves transaction changes
ROLLBACK	Undoes uncommitted changes
SAVEPOINT	Sets a rollback checkpoint

SQL CONSTRAINTS

Rules for data accuracy and integrity

PRIMARY KEY Uniquely identifies each record in a table (NOT NULL + UNIQUE).	FOREIGN KEY Establishes a link/relationship between data in two tables.
UNIQUE Ensures all values in a specified column are distinct.	NOT NULL Prevents NULL (empty) values from being stored in a column.